113 學年度

國立中山大學

課程名稱

verilog

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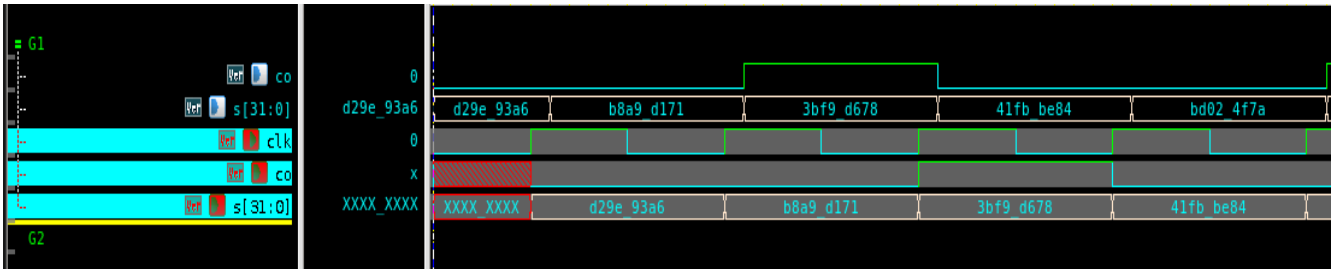
III.

RTL Wave Form:

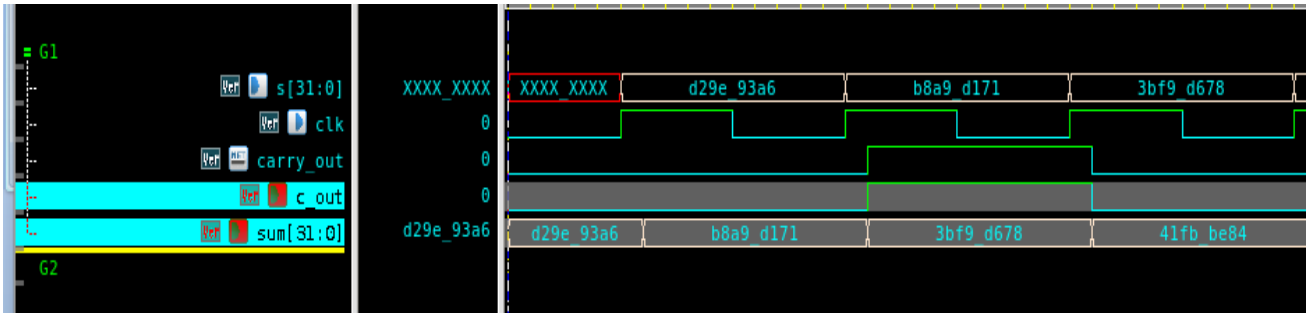


Gate Level Wave Form:

Behavior:



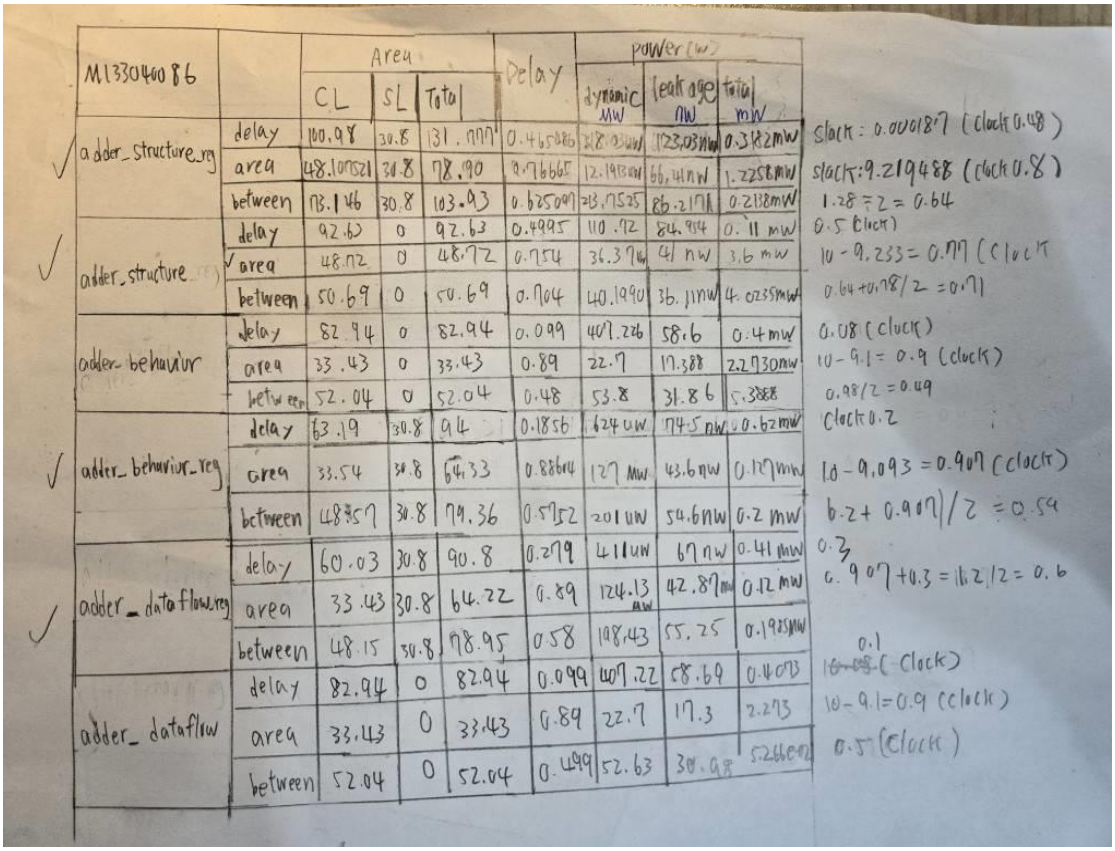
Structure:



DataFlow:



IV.數據表格:



M1330460086		Area CL	Area SL	Area Total	Delay	power dynamic	power leakage	power total
adder_structure_reg	delay	100.98	30.8	131.7777	0.465886	328.31uW	123.03nW	0.342mW
	area	48.10852	30.8	78.190	0.76665	12.198uW	66.41nW	1.2258mW
	between	73.146	30.8	103.93	0.625097	73.7525	86.217n	0.2138mW
adder_structure	delay	92.63	0	92.63	0.4995	110.72	84.954	0.11 mW
	area	48.72	0	48.72	0.754	36.376	41 nW	3.6 mW
	between	50.69	0	50.69	0.704	40.1990	36.11nW	4.0235mW
adder_behavior	delay	82.94	0	82.94	0.099	407.226	58.6	0.4mW
	area	33.43	0	33.43	0.89	22.7	17.388	2.2930mW
	between	52.04	0	52.04	0.48	53.8	31.86	5.3888
adder_behavior_reg	delay	63.19	30.8	94	0.1856	624uW	74.5nW	0.62mW
	area	33.54	30.8	64.33	0.88614	127 MW	43.6nW	0.17mW
	between	48.57	30.8	79.36	0.5752	201uW	54.6nW	0.2 mW
adder_dataflow_reg	delay	60.03	30.8	90.8	0.279	411uW	67nW	0.41mW
	area	33.43	30.8	64.22	0.89	124.13 AW	42.87nW	0.12mW
	between	48.15	30.8	78.95	0.58	198.43	55.25	0.1925mW
adder_dataflow	delay	82.94	0	82.94	0.099	407.22	58.69	0.4073
	area	33.43	0	33.43	0.89	22.7	17.3	2.273
	between	52.04	0	52.04	0.499	52.63	30.97	5.266mW

slack: 0.000187 (clock 0.48)
slack: 9.219488 (clock 0.8)
1.28 ÷ 2 = 0.64
0.5 (clock)
10 - 9.233 = 0.767 (clock
0.64 + 0.78/2 = 0.71
0.08 (clock)
10 - 9.1 = 0.9 (clock)
0.98/2 = 0.49
clock 0.2
10 - 9.093 = 0.907 (clock)
0.2 + 0.907)/2 = 0.54
0.3
0.907 + 0.3 = 1.2/2 = 0.6
0.1
(clock)
10 - 9.1 = 0.9 (clock)
0.5 (clock)

Area/Delay/Between比較數據&截圖report 存在檔案”HW1截圖”中

V.

有 reg 的會比較慢一點才有質因為在等 clk 才會將質存入 reg

VI.心得:

做完數據表格有發現 delay optimize 的 arrival time 會比較小但相對的 area optimize 就會比較大有種用空間換取時間的概念 area optimize 則相反。power 功率消耗是 area optimize 比較高,推測電路大功率消耗也大,但 power leakage 則是 delay optimize 比較高。這次作業因為是第一次接觸 verilog 所以遇到的問題很多,也很感謝耐心回復我信件的助教,幫了我非常多忙。